Probability Topics

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Question 1

0/1 pt 3 19

Giving a test to a group of students, the grades and genders are summarized below.

	A	B	C	Total
Male	14	15	5	34
Female	13	19	3	35
Total	27	34	8	69

If one student is chosen at random,

find the probability that the student was male OR received a grade of "C". (Leave answer to four decimal places.)

Question Help: Video

Question 2

0/1 pt 3 19

If you randomly select a card from a well-shuffled standard deck of 52 cards, what is the probability that the card you select is a heart or diamond? (Your answer must be in the form of a reduced fraction.)

Question Help: Video 1 Video 2

Question 3

0/1 pt 3 19

A group of people were asked if they had run a red light in the last year. 151 responded "yes", and 263 responded "no".

Find the probability that if a person is chosen at random, they have run a red light in the last year.

Give your answer as a fraction or decimal accurate to at least 3 decimal places

Question Help: Video 1 Video 2

Question 4

0/1 pt 3 19

Giving a test to a group of students, the grades and gender are summarized below

	A	B	C	Total
Male	14	16	15	45
Female	7	12	10	29
Total	21	28	25	74

If one student is chosen at random,

Find the probability that the student did NOT get an "B"

Question Help: Video 1 Video 2

Question 5

0/1 pt 3 19

A group of people were asked if they owned a dog. 199 responded "yes", and 390 responded "no".

Find the probability that if a person is chosen at random, they own a dog.

- $\frac{191}{199}$
- $\frac{199}{589}$
- $\frac{199}{390}$

Question Help: Video

Question 6

0/1 pt 3 19

An American roulette wheel has 38 slots: two slots are numbered 0 and 00, and the remaining slots are numbered from 1 to 36. Find the probability that the ball lands in an odd-numbered slot, not including 0 or 00.

Your answer is :

Question Help: Video

Question 7

0/1 pt 3 19

In a study of brand recognition, 944 consumers knew of Hershey, and 4 did not. Use these results to estimate the probability that a randomly selected consumer will recognize Hershey.

Report your answer as a decimal accurate to 3 decimal places.

prob =

Question Help: Video

Question 8

0/1 pt 3 19

How many outcomes would there be in the sample space for rolling 2 dice and flipping 5 coins?

Question 9

0/1 pt 3 19

Events A and B are indpendent events. Find the indicated Probability.

`P(A) = 0.5`

`P(B) = 0.4`

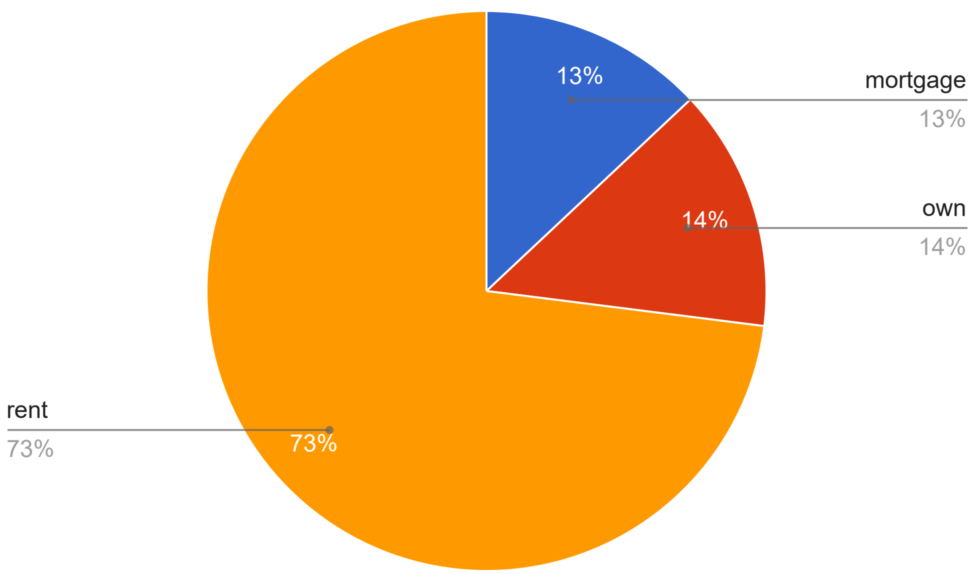
`P( A and B) =`

Question Help: Video

Question 10

0/1 pt 3 19

A summary of home ownership statuses of loan applicants is displayed in the pie chart below.



Suppose a loan is selected at random.

What is the probability that the applicant owns his/her home?

What is the probability that the applicant rents?

What is the probability that the applicant does not have a mortgage?

Question Help: Video

Question 11

0/1 pt 3 19

The following contingency table summarizes the school enrollment by level and type:

	Public (`T1`)	Private (`T2`)	Total
Elementary School (`L1`)	375	75	____
Middle School (`L2`)	180	50	____
High School (`L3`)	196	99	____
Total	____	____	____

a. For a randomly selected student, describe the following in words:
- `L2` is Select an answer
- `P(L2)` is Select an answer

b. Find the following probabilities: (Round the answers to 4 decimal places)

i. `P(L2)=` ____

ii. `P(T1)=` ____

iii. `P(L2 cap T2)=` ____

iv. `P(L1 cupT2)=` ____

v. `P(L2 | T1 )=` ____

c. Answer the following questions:

i. Are `L3` and `T2` mutually exclusive? Select an answer Explain.
Select an answer

ii. Are `L3` and `T2` independent? Select an answer Explain.
Select an answer

iii. Are `L2` and `L3` mutually exclusive? Select an answer Explain.
Select an answer

iv. Are `L2` and `L3` independent? Select an answer Explain.
Select an answer

Question Help: Video

Question 12

0/1 pt 3 19

The following contingency table summarizes the relation between whether a millennial attended a college or not and the number of their parents that completed a college degree:

	attended (`C1`)	didn't attend (`C2`)	Total
0 (`E1`)	215	265	______
1 (`E2`)	252	65	______
2 (`E3`)	272	20	______
Total	______	______	______

a. For a randomly selected student, describe the following in words:
- `C2` is Select an answer
- `P(C2)` is Select an answer

b. Find the following probabilities: (Round the answers to 4 decimal places)

i. `P(E3)=` ______

ii. `P(C1)=` ______

iii. `P(E2 cap C1)=` ______

iv. `P(E2 cupC2)=` ______

v. `P(E1 | C1 )=` ______

c. Answer the following questions:

i. Are `E1` and `C1` mutually exclusive? Select an answer Explain.
Select an answer

ii. Are `E1` and `C1` independent? Select an answer Explain.
Select an answer

iii. Are `E2` and `E3` mutually exclusive? Select an answer Explain.
Select an answer

iv. Are `E2` and `E3` independent? Select an answer Explain.
Select an answer

Question Help: Video 1 Video 2

Question 13 0/1 pt 3 19

Quetta has surveyed their 27 classmates about the number of jobs a person held in their lifetime and received the following frequency table:

Number	Frequency
0	1
1	6
2	5
3	5
4	4
5	2
6	4

Find the probability that for a randomly selected classmate the number of jobs a person held in their lifetime is 6.

______ (Round the answer to 4 decimal places.)

Question Help: Video

Question 14 0/1 pt 3 19

Find the probability that there is one T when a fair coin is flipped three times.

______ % (Round the answer to 2 decimal points.)

Question Help: Video

Question 15 0/1 pt 3 19

Find the probability that there are no H when a fair coin is flipped two times.

______ % (Round the answer to 2 decimal points.)

Question Help: Video

Question 16 0/1 pt 3 19

Jayceon measured how many times he arrived to work late if he took subway. The results are provided in the following table:

Result	Frequency
late	5
on-time	21

Find the probability that Jayceon will arrive on-time tomorrow if he plans to take subway.

(Round the answer to 4 decimal places.)

Question Help: Video

Question 17 0/1 pt 3 19

Let E be the event defined as { 2 3 }. The result of rolling a die is 5. Did E occur or not?

?

Question Help: Video

Question 18 0/1 pt 3 19

Let's roll a die and let E be the event defined as { 1 3 6 }. Find P(E).

(Round the answer to 4 decimals.)

Question Help: Video

Question 19 0/1 pt 3 19

Let `S` be the sample space of drawing a random number from 1 to 20, so that:

`S = {1,2,3,...,18,19,20}`

Let `A` and `B` be events, where:

Event `A = \{1,2,3,5,6,8,9,10,11,16,17,18,20}`

Event `B = \{3,5,6,8,9,10,13,14,16,19}`

List all simple outcomes that define the event (A or B):
(A or B) = { ____ }
Enter the simple outcomes as a list, separated by commas. If the result is the empty set, enter **DNE**

List all simple outcomes that define the event (A and B):
(A and B) = { ____ }
Enter the simple outcomes as a list, separated by commas. If the result is the empty set, enter **DNE**

List all simple outcomes that define the event (not A):
(not A) = { ____ }
Enter the simple outcomes as a list, separated by commas. If the result is the empty set, enter **DNE**

Question Help: Video

Question 20 0/1 pt 3 19

Let `A` be an event, such that `P(A)=0.1998` . Find the following probability:

P(not A)= ____ % (Round the answer to 2 decimals)

Question Help: Video

Question 21 0/1 pt 3 19

Let `A` and `B` be mutually exclusive events, such that `P(A)=0.6066` and `P(B)=0.0948` . Find the following probabilities:

P(A and B)= ____ (Round the answer to 4 decimals)

P(A or B)= ____ (Round the answer to 4 decimals)

Question Help: Video

Question 22 0/1 pt 3 19

Let `A` and `B` be two events, such that `P(A)=0.292`, `P(B)=0.2848`, and `P(A and B)=0.1427`. Find the following probability:

P(A or B)= ________ % (Round the answer to 2 decimals)

Question Help: Video

Question 23

0/1 pt 3 19

Let `A` and `B` be two events, such that `P(A)=0.59` and `P(B|A)=0.51`. Find the following probability:

P(A and B)= ________ (Round the answer to 4 decimals)

Question Help: Video

Question 24

0/1 pt 3 19

Let `A` and `B` be independent events, such that `P(A)=0.7068` and `P(B)=0.6989` . Find the following probabilities:

P(A and B)= ________ % (Round the answer to 2 decimals)

P(A or B)= ________ % (Round the answer to 2 decimals)

Question Help: Video 1 Video 2

Question 25

0/1 pt 3 19

Jack has surveyed 25 classmates about the number of jobs a person held in their lifetime and received the following frequency table:

Number	Frequency
0	8
1	4
2	1
3	4
4	2
5	4
6	2

Let A be the event in which for a randomly selected classmate the number of jobs a person held in their lifetime is 2 and let B be the event in which for a randomly selected classmate the number of jobs a person held in their lifetime is 5.

Find the following probabilities:

P(A or B) = ________ (Round the answer to 4 decimal places.)

P(A and B) = ________ (Round the answer to 4 decimal places.)

Question Help: Video

Question 26 0/1 pt 3 19

The following contingency table summarizes the drug test results for 441 test-takers:

	Positive	Negative	Total
Doesn't use drugs	6	184	____
Uses drugs	236	15	____
Total	____	____	____

For a randomly selected test administration, find the following probabilities. (Round the answers to 4 decimal places.)

1) The probability that the test result is negative:

____

2) The probability of a true positive:

____

3) The probability of a correct inference:

____

Question Help: Video

Question 27

0/1 pt 3 19

The following contingency table summarizes the drug test results for 405 test-takers:

	Positive	Negative	Total
Doesn't use drugs	1	152	
Uses drugs	241	11	
Total			

For a randomly selected test administration, find the following probabilities. (Round the answers to 4 decimal places.)

1) The probability that the test-taker uses drugs:

2) The probability of a true positive:

3) The probability of an error:

4) The negative predicting value:

Question Help: Video 1 Video 2

Question 28

0/1 pt 3 19

Let `A` and `B` be independent events, such that `P(A)=0.1093` and `P(B)=0.0408` . Find the following probabilities:

P(A and B)= % (Round the answer to 2 decimals)

Question Help: Video

Question 29

0/1 pt 3 19

Let `A` and `B` be mutually exclusive events, such that `P(A)=0.5667` and `P(B)=0.1894` . Find the following probabilities:

P(A or B)= (Round the answer to 4 decimals)

Question Help: Video

Question 30 0/1 pt 3 19

In the first few parts, identify whether the events are Mutually Exclusive, Independent, or Neither (events cannot be both disjoint and independent).

a. You and a randomly selected student from your class both earn A's in this course.

- Independent
- Mutually Exclusive
- Neither

b. You and your class partner both earn A's in this course.

- Neither
- Mutually Exclusive
- Independent

c. While driving your car to school you could either crash or not.

- Mutually Exclusive
- Independent
- Neither

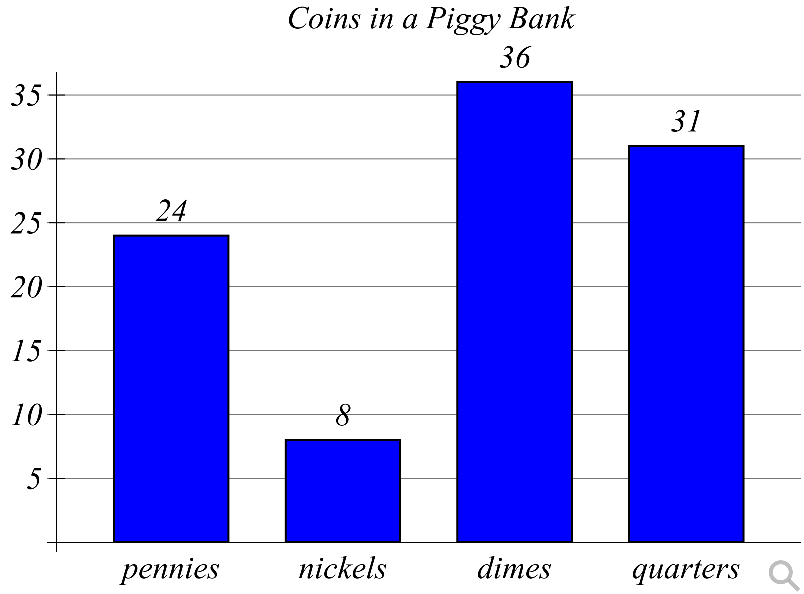
d. If two events can occur at the same time, they must be independent.

- False
- True

Question Help: Read

Question 31 0/1 pt 3 19

The following bar graph shows the distribution of pennies, nickels, dimes, and quarters in a piggy bank.



Suppose a coin is selected at random from the piggy bank. Answer the problems below to 2 decimal places.

What is the probability that the coin is a penny?

What is the probability that the coin is a dime?

What is the probability that the coin is not a nickel?

Question Help: Video

Question 32 0/1 pt 3 19

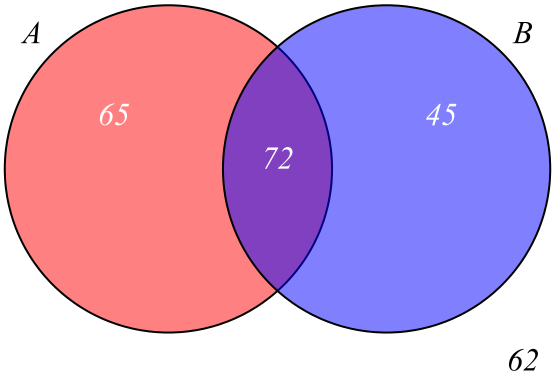
A coin will be tossed twice. Let `E` be the event "the first toss shows heads" and `F` the event "the second toss shows heads".

(a) Are the events `E` and `F` independent?
Input Yes or No here:

(b) Find the probability of showing heads on both tosses.
Input your answer here:

Question Help: Video

Question 33 0/1 pt 3 19



Answer the following using the Venn Diagram:

How many items are in set `A`? __________

How many items are in set `B`? __________

How many items are represented in the Venn Diagram overall? __________

What is the probability of selecting an item at random from set `A`? __________

Question Help: Video